

SARTHAK AJAY PATIL

EDUCATION

Indian Institute of Technology (BHU), Varanasi <i>Integrated Dual Degree in Civil Engineering</i>	<i>2023 - 2028</i> <i>8.31</i>
Arihant College <i>Maharashtra State Board of Secondary and Higher Secondary Education</i>	<i>2023</i> <i>75.50%</i>
Priyadarshani School <i>Central Board of Secondary Education</i>	<i>2021</i> <i>92.00%</i>

INTERNSHIPS

Mapping of Glaciers using Deep Learning Techniques <i>Summer Research Intern under Dr. Saurabh Vijay</i>	<i>May 2025 - July 2025</i> <i>IIT Roorkee</i>
- Developed DeepLabV3+, ResUNet and SegFormer-B4 pipelines for glacier segmentation using PyTorch and TensorFlow. - Engineered a preprocessing pipeline for 7,019 18-band image patches using normalization and data augmentation. - Optimized model training with a hybrid BCE, Dice, IoU and SSIM loss for accurate glacier boundary prediction. - Benchmarked four deep learning segmentation models across 5 regions, achieving IoU 0.897 and F1-score 0.946. - Performed comparative analysis using IoU, Accuracy, Precision, Recall, F1 and Kappa to select the best architecture.	

SKILLS

- **Programming Languages:** C, C++, Python, JavaScript, SQL
- **Backend and Web:** FastAPI, React.js, REST APIs, SQLAlchemy, JWT Authentication
- **Databases and Caching:** PostgreSQL, Redis
- **CS Fundamentals:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks.
- **ML Libraries:** PyTorch, TensorFlow, Scikit Learn, Pandas, NumPy, Matplotlib

PROJECTS

URL Shortener <i>FastAPI, PostgreSQL, Redis, Docker</i>
• Built a URL-shortening service with short-code generation, custom aliases, and click analytics behind secure REST APIs. • Implemented a cache-aside Redis layer over indexed PostgreSQL lookups, reducing redirect latency for repeated requests. • Designed JWT authentication, rate limiting, and input validation to secure and harden the public API surface. • Optimized PostgreSQL schema and REST APIs for efficient URL storage and retrieval; Dockerized the stack.
Data Engineering Copilot <i>TechStack: FastAPI, React, PostgreSQL, Pandas, SQLAlchemy, Plotly, Docker</i>
• Designed a scalable FastAPI backend with modular services for automated dataset profiling workflows. • Implemented secure file upload, validation, metadata persistence and REST APIs using PostgreSQL. • Developed reusable data-processing pipelines for schema inference, statistics and quality analysis. • Built a React dashboard with Plotly to visualize profiling reports and dataset insights.

POSITIONS OF RESPONSIBILITY

Mentor ML Cohort Research Cell
• Created a ML learning roadmap covering Python, Machine Learning, Deep Learning and project-based learning. • Mentored students in Machine Learning, resolving conceptual and implementation challenges across core ML topics.
Operations Head Kashiyaatra'26
• Directed operations for 50+ events with 25,000+ attendees, ensuring flawless infrastructure and smooth execution. • Led 40+ coordinators, efficiently managing logistics, vendors, operations, and seamless cross-team collaboration.

ACHIEVEMENTS

- Solved **500+ DSA problems** across all platforms, strengthening data structures and algorithmic skills.
- **National Semi-Finalist**, Flipkart GRID 7.0, selected among teams nationwide in Flipkart's flagship SDE challenge.
- **Achieved AIR 1371** among thousands of participants in Coding Ninjas Ninja Slayground 2.0 coding challenge.
- **Finalist**, EcoVision Sustainability Strategy Challenge conducted by Advaita'24 in collaboration with ISB and FundVice

EXTRA CURRICULARS

- Enhanced knowledge of **software engineering** and **development** workflows through COPS Week'25 at IIT (BHU).
- Competed in the Kuku FM **Chess Tournament**, demonstrating competitive spirit and disciplined strategic gameplay.